

Model: Hongik Evacuation

Description: There is a two-storey building in this model. On the first floor, of this 24 person office building, the workers work and socialize. The restroom is also located on this floor. Workers typical work day consists of eight hours.

On the second floor there is a cafeteria, where workers and outside customers have lunch. (The number of outside customers usually depends what time of day it is – there will be more during lunch than in the morning for breakfast)

There is only one stairway between the two floors and one or two exits from the first floor.

When the fire alarm is activated everyone in the building will go to one of the exits. Some may not react immediately and keep working while others may freeze or panic. Some may also want to get back to their tables before exiting the building.

The alarm will start at the specified time or by pressing the "Fire alarm" button in the simulation window.

Name	Value
General	
Model time units	minutes
System Dynamics solver	
Differentiation Equations Method	Euler
Algebraic Equations Method	Modified Newton
Mixed Equations Method	RK45+Newton
Absolute accuracy	1.0E-5
Time accuracy	1.0E-5
Relative accuracy	1.0E-5
Fixed time step	0.001
Advanced	
Java package name	evacuation
File Name	/Users/yschung/Models/Hongik Evacuation/Hongik Evacuation.alp
Description	
Description	There is a two-storey building in this model. On the first floor, of this 24 person office building, the workers work and socialize. The restroom is also located on this floor. Workers typical work day consists of eight hours. On the second floor there is a cafeteria, where workers and outside customers have lunch. (The number of outside customers usually depends what time of day it is – there will be more during lunch than in the morning for breakfast) There is only one stairway between the two floors and one or two exits from the first floor. When the fire alarm is activated everyone in the building will go to one of the exits. Some may not react immediately and keep working while others may freeze or panic. Some may also want to get back to their tables before exiting the building. The alarm will start at the specified time or by pressing the "Fire alarm" button in the simulation window.

Agent Type: Main

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true

Name	Value
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Pedestrian library objects

- pedConfiguration
- pedAreaStairwayUp
- pedAreaStairwayDown

Auxiliary functions

- getPeopleInsideCount
- getEvacuationTime

Arrival rates

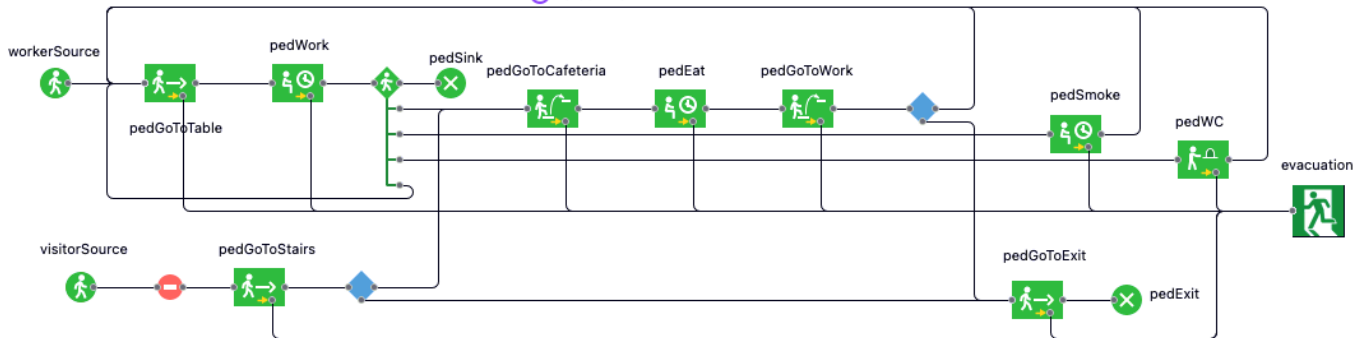
- visitorArrivalMultiplier
- visitorsArrivalRate
- workerArrivalRate

Animation

- timeToString

Timing

- alarmAuto
- alarmTime
- addExit
- alarm
- alarmBeginning
- alarmEnding
- evacuationStart
- evacuationEnds
- closeCafeteria
- cafeteriaOpen
- workTables
- visitorCount
- evacTime



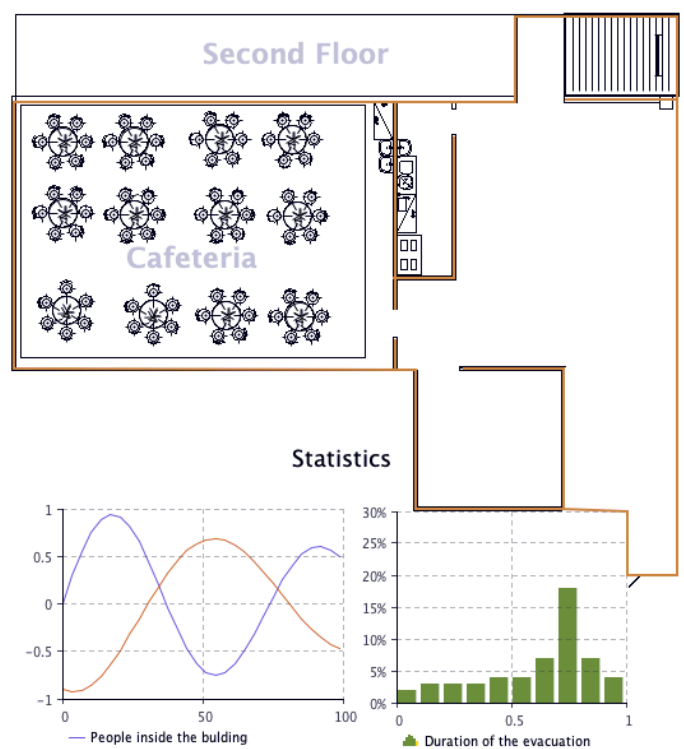
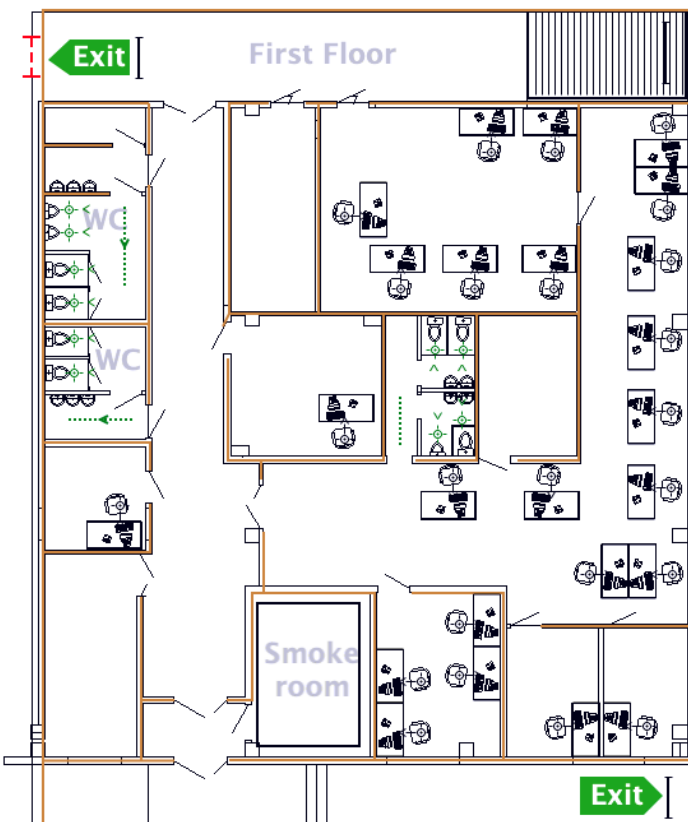
Hongik Evacuation model

Evacuation time: 00:00

People inside the building: 0

Additional exit: ☒

Time: 00:00

Flood Alarm!


Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	50.0
y	-690.0
Rotation	0.0

Parameter: alarmTime

Name	Value
General	
Array	false
Default value	540
Type	double
Show at runtime	true
Show name	true
Value editor	
Label	Alarm time [min]
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: addExit

Name	Value
General	
Array	false
Default value	true
Type	boolean
Show at runtime	true
Show name	true
Value editor	
Label	Add exit
Editor control	Check Button
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: visitorArrivalMultiplier

Name	Value
General	
Array	false
Default value	2
Type	double
Show at runtime	true
Show name	true
Value editor	
Label	Visitor arrival multiplier
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Parameter: workerArrivalRate

Name	Value
General	
Array	false
Default value	(60 : PER_HOUR)
Unit	per hour
Show at runtime	true
Show name	true
Value editor	
Label	Worker arrival rate
Editor control	Unit editor
Advanced	
System dynamics units	false
Save in snapshot	true

Function: getPeopleInsideCount

Name	Value
General	
Return type	long
Return type:	Returns value
Show at runtime	true
Show name	true
Function body	
Body	return workerSource.countPeds() + visitorCount - pedSink.sink.count() - pedExit.sink.count() - evacuation.pedSink.sink.count();
Advanced	
Access type	default
System dynamics units	false

Function: getEvacuationTime

Name	Value
General	
Return type	double

Name	Value
Return type:	Returns value
Show at runtime	true
Show name	true
Function body	
Body	return (alarmEnding == 0 ? (time() - alarmBeginning)/second() : (alarmEnding - alarmBeginning)/second());
Advanced	
Access type	default
System dynamics units	false

Function: timeToString

Name	Value
General	
Return type	String
Return type:	Returns value
Show at runtime	true
Show name	true
Function body	
Body	Date date = timeToDate(minutes); int hour = getHourOfDay(date); int minute = getMinute(date); String hourToString = hour < 10 ? "0" + hour : hour + ""; String minuteToString = minute < 10 ? "0" + minute : minute + ""; return hourToString + ":" + minuteToString;
Advanced	
Access type	default
System dynamics units	false

Arguments:

Name	Type
minutes	double

Schedule: visitorsArrivalRate

Name	Value
General	
Show at runtime	true
Show name	true
Data	
Value type	real
The schedule defines	Intervals (Start, End)
Representation type	Days/Weeks
Repeat time	1
Repeat time interval	days
Is snapped to date	false
Default value	0.0
Load From Database	false
Preview	
SCHEDULE_PREVIEW_START_DATE	1762897960288
Advanced	

Name	Value
System dynamics units	false

Schedule intervals:

Start	End	Value
8:00 AM	11:00 AM	0.2
11:00 AM	2:00 PM	0.5
2:00 PM	4:00 PM	0.33
4:00 PM	10:00 PM	0.25

Event: evacuationEnds

Name	Value
General	
Logging	true
Condition	alarm && getPeopleInsideCount() == 0
Trigger type	Condition
Show at runtime	true
Show name	true
Action	
Action	getEngine().finish(); alarmEnding = time();

Event: alarmAuto

Name	Value
General	
Logging	true
EVENT_TIMEOUT_PROPERTIES	- Recurring Event Properties
Mode	Occurs once
Trigger type	Timeout
Show at runtime	true
Show name	true
Action	
Action	button.action();

Event: closeCafeteria

Name	Value
General	
Logging	true
EVENT_TIMEOUT_PROPERTIES	- Recurring Event Properties
Mode	Occurs once
Trigger type	Timeout
Show at runtime	true
Show name	true
Action	
Action	cafeteriaOpen = false; pedEat.freeAll();

Variable: alarm

Name	Value
General	
Initial value	false
Type	boolean
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: alarmBeginning

Name	Value
General	
Initial value	0
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: alarmEnding

Name	Value
General	
Initial value	0
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: cafeteriaOpen

Name	Value
General	
Initial value	true
Type	boolean
Show at runtime	true
Show name	true

Name	Value
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: visitorCount

Name	Value
General	
Initial value	0
Type	int
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Collection: workTables

Name	Value
General	
Initial contents	{ seat0, seat1, seat2, seat3, seat4, seat5, seat6, seat7, seat8, seat9, seat10, seat11, seat12, seat13, seat14, seat15, seat16, seat17, seat18, seat19, seat20, seat21, seat22, seat23 }
Initial contents	{ seat0, seat1, seat2, seat3, seat4, seat5, seat6, seat7, seat8, seat9, seat10, seat11, seat12, seat13, seat14, seat15, seat16, seat17, seat18, seat19, seat20, seat21, seat22, seat23 }
Element class	Attractor
Collection class	ArrayList
Show at runtime	true
Show name	true
Advanced	
Access type	public
Save in snapshot	true

Evacuation: evacuation

Name	Value
General	
Population of agents	false
Population of agents	false
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
exits	<pre>new ArrayList<TargetLine>(){ add(enter); if (addExit) add(line30); }</pre>

PedSource: workerSource

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Appears at	self.LOCATION_LINE
Target line	enter
Arrive according to	self.RATE
Arrival rate	workerArrivalRate
Limited number of arrivals	true
Maximum number of arrivals	workTables.size()
New pedestrian	new OfficeWorker(workTables.get((int)self.countPeds()))
Comfortable speed	uniform(0.5, 1)
Initial speed	uniform(0.3, 0.7)
Diameter	uniform(0.4, 0.5)
Create groups	false
Add pedestrian to:	false
On exit	ped.isSmoker = randomTrue(0.3);

PedSink: pedSink

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

PedGoTo: pedGoToStairs

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_LINE
Target line	line20
Reach tolerance	0.25
On enter	ped.person.setVisible(true); visitorCount++;

SelectOutput: selectOutput

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Select True output	false
Condition	cafeteriaOpen
Condition is stochastic or volatile	true

PedSource: visitorSource

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Agent]
Population of agents	false

Name	Value
Generic Parameters Substitutes	[Generic Parameter Substitute: Agent]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Appears at	self.LOCATION_LINE
Target line	enter
Arrive according to	self.RATE_SCHEDULE
Rate schedule	visitorsArrivalRate
Modify rate	true
Rate expression	baseRate * visitorArrivalMultiplier
Limited number of arrivals	false
New pedestrian	new evacuation.Visitor()
Comfortable speed	uniform(0.5, 1)
Initial speed	uniform(0.3, 0.7)
Diameter	uniform(0.4, 0.5)
Create groups	false
Add pedestrian to:	false

PedWait: pedEat

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Waiting location	self.LOCATION_NODE
Node	cafeteria
Attractor choice	self.ATTRACTOR_CHOICE_FREE
Delay ends	self.DELAY_TIMEOUT
Delay time	uniform(0.5, 1.0)
Group members leave together	false
Delay starts when	self.DELAY_STARTS_SINCE_BEGIN_WAIT
Maximum capacity	true
Reach tolerance	0.25

PedGoTo: pedGoToExit

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_LINE
Target line	enter
Reach tolerance	0.25

PedSink: pedExit

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

PedWait: pedSmoke

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Agent Parameters:

Name	Value
Waiting location	self.LOCATION_NODE
Node	smokeRoom
Attractor choice	self.ATTRACTOR_CHOICE_NONE
Delay ends	self.DELAY_TIMEOUT
Delay time	uniform(5,25)
Group members leave together	false
Delay starts when	self.DELAY_STARTS_SINCE_BEGIN_WAIT
Maximum capacity	true
Reach tolerance	0.25

PedService: pedWC

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Services	servicesWC
Queue choice policy	self.CHOICE_CLOSEST_QUEUE
Delay time	uniform(2.0, 3.0)
Recovery delay	1
Pass through in reverse direction	false

PedSelectOutput: pedSelectOutput

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
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Agent Parameters:

Name	Value
Use:	self.TYPE_CONDITIONS
Condition 1	toTimeUnits(ped.worktime, HOUR) > 7*hour()
Condition 2	ped.isHungry && toTimeUnits(ped.worktime, HOUR) > 2*hour() && cafeteriaOpen
Condition 3	ped.isSmoker && randomTrue(0.4)
Condition 4	randomTrue(0.5)
Choice is stochastic or volatile	true
On exit 2	ped.isHungry = false;

PedGoTo: pedGoToTable

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: OfficeWorker]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_POINT
X	ped.table.getX()
Y	ped.table.getY()
Reach tolerance	0.25

PedAreaDescriptor: pedAreaStairwayUp

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Area	stairwayUp

Agent Parameters:

Name	Value
Speed is limited	false
Speed is multiplied	true
By factor of	0.5
Is moving	false
Enable access control	false
Throughput	+infinity

PedSettings: pedConfiguration

Name	Value
General	
Population of agents	false
Population of agents	false
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Enable social distancing	false
Allow pedestrian selection by click	false
Time step	0.3
Density modifier	10
Pedestrian mass	120
Routing touch tolerance	1.25

PedAreaDescriptor: pedAreaStairwayDown

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Area	stairwayDown
Speed is limited	false
Speed is multiplied	true
By factor of	0.7

Agent Parameters:

Name	Value
Is moving	false
Enable access control	false
Throughput	+infinity

PedWait: pedWork

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Waiting location	self.LOCATION_NODE
Node	workArea
Attractor choice	self.ATTRACTOR_CHOICE_CUSTOM
Attractor	ped.table
Delay ends	self.DELAY_TIMEOUT
Delay time	triangular(0.5,0.7,1)
Group members leave together	false
Delay starts when	self.DELAY_STARTS_SINCE_BEGIN_WAIT
Maximum capacity	true
Reach tolerance	0.25
On enter	ped.worktime += self.delayTimeout(ped) ;

Hold: hold

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
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Agent Parameters:

Name	Value
Mode	self.CONDITIONAL
Blocking condition	alarm
Initially blocked	false

PedChangeLevel: pedGoToCafeteria

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Exit line on current level	line20
Entry line on new level	line21
Reach tolerance	0.25

PedChangeLevel: pedGoToWork

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Exit line on current level	line21
Entry line on new level	line20
Reach tolerance	0.25

SelectOutput: selectOutput1

Name	Value
General	

Name	Value
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Select True output	false
Condition	agent instance of OfficeWorker
Condition is stochastic or volatile	true

Time Plot: plot

Name	Value
General	
Lock	false
Public	true
Data update	
Analysis auto update	true
ANALYSIS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Dataset Samples To Keep	100
Scale	
Time window	100
Time	model time units
Vertical scale	Auto
Appearance	
Labels horizontal position	DEFAULT
Labels vertical position	DEFAULT
Label format	Model time units
Labels Text Color	darkGray
Chart Area Grid Color	darkGray
Draw line	true
Fill area under line	false
Interpolation	Linear
Position and size	
x	570.0
Width	280.0
y	490.0
Height	250.0
Legend	
Show legend	true
Legend size	61.0
Legend text color	black
Chart area	

Name	Value
Chart Area: X Offset	40.0
Chart Area: Width	220.0
Chart Area: Y Offset	20.0
Chart Area: Height	149.0
Chart Area: Background Color	white
Chart area border color	black
Omniverse	
Usd binding type	None
Advanced	
Time window moves	As data becomes available
Show name	false
Logging	true

Plot Items:

Title	Type	Dataset / Value	Point Style	Color	Line	Width	Interpolation
People inside the bulding	value	getPeopleInsideCount()	NONE	slateBlue	true	1.0	LINEAR
Begining of the evacuation	dataset	evacuationStart	NONE	chocolate	true	1.0	LINEAR

Data Set: evacuationStart

Name	Value
General	
Dataset Samples To Keep	100
Axis Data Freeze X Axis	false
Show at runtime	true
Show name	true
Data update	
Analysis auto update	false
Logging	true

Text: text

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	0
Dynamic: Text	getPeopleInsideCount()
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	550.0
y	50.0

Name	Value
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text1

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	People inside the building:
Appearance	
Color	black
Alignment	RIGHT
Position and size	
x	540.0
y	50.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text2

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	00:00
Dynamic: Text	alarm ? (((int) getEvacuationTime())/60 + ":" + (((int) getEvacuationTime())%60 < 10 ? "0" : "") + ((int)(alarmEnding == 0 ? (time() - alarmBeginning)/second() : (alarmEnding - alarmBeginning)/second()))%60) : "00:00"
Appearance	
Color	navy
Alignment	LEFT
Position and size	

Name	Value
x	190.0
y	50.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text3

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Evacuation time:
Appearance	
Color	black
Alignment	RIGHT
Position and size	
x	170.0
y	50.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Rectangle: rectangle1

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	30.0
Width	520.0

Name	Value
y	120.0
Height	640.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Polyline: polyline

Name	Value
General	
Polyline closed	false
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	573.0
y	202.0
z	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Rectangle: rectangle

Name	Value
General	
Dynamic: Visible	checkbox.isSelected()
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Fill color	white
Line width	1.0
Line style	solid
Position and size	

Name	Value
x	19.0
Width	41.0
y	140.0
Height	31.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text4

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	First Floor
Appearance	
Color	silver
Alignment	LEFT
Position and size	
x	200.0
y	140.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text5

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Second Floor
Appearance	
Color	silver

Name	Value
Alignment	LEFT
Position and size	
x	720.0
y	140.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Agent Presentation: evacuation_presentation

Name	Value
General	
Show at runtime	true
Public	true
Position and size	
x	752.0
y	212.0
z	0.0
Rotation	0.0
Scale	Automatically calculated
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text6

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Statistics
Appearance	
Color	black
Alignment	LEFT
Position and size	
x	790.0
y	460.0
z	0.0
Rotation	0.0
Omniverse	

Name	Value
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text7

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	WC
Appearance	
Color	silver
Alignment	LEFT
Position and size	
x	70.0
y	270.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text9

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Smoke room
Appearance	
Color	silver
Alignment	CENTER
Position and size	
x	250.0
y	610.0
z	0.0
Rotation	0.0
Omniverse	

Name	Value
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text11

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Additional exit:
Appearance	
Color	black
Alignment	RIGHT
Position and size	
x	180.0
y	80.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
On click	checkbox.setSelected(!checkbox.isSelected(),true);
Show name	false

Text: text12

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Cafeteria
Dynamic: Text	"Cafeteria" + (cafeteriaOpen ? "" : " (closed)")
Appearance	
Color	silver
Alignment	LEFT
Position and size	
x	660.0
y	300.0
z	0.0
Rotation	0.0

Name	Value
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text13

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Pedestrian library objects
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	40.0
y	-580.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line72

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	40.0
dX	190.0
y	-560.0
dY	0.0
z	0.0
dZ	0.0

Name	Value
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text14

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Auxiliary functions
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	260.0
y	-580.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line73

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	260.0
dX	160.0
y	-560.0
dY	0.0
z	0.0

Name	Value
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text15

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Timing
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	440.0
y	-580.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line85

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	440.0
dX	140.0
y	-560.0
dY	0.0

Name	Value
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line86

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	260.0
dX	160.0
y	-480.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text16

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Arrival rates
Appearance	
Color	navy
Alignment	LEFT
Position and size	

Name	Value
x	260.0
y	-500.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line87

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	260.0
dX	160.0
y	-380.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text17

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Animation
Appearance	
Color	navy
Alignment	LEFT

Name	Value
Position and size	
x	260.0
y	-400.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text8

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	WC
Appearance	
Color	silver
Alignment	LEFT
Position and size	
x	80.0
y	380.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line90

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	25.0

Name	Value
dX	10.0
y	171.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line91

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	25.0
dX	10.0
y	140.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text20

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	Time:
Appearance	

Name	Value
Color	black
Alignment	RIGHT
Position and size	
x	540.0
y	80.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text21

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Text	
Text	00:00
Dynamic: Text	timeToString(time())
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	550.0
y	80.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Group: groupFloor1

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	279.0

Name	Value
y	440.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

CAD Drawing: cad

Name	Value
General	
Invert Default Colors	true
Fill color	white
File Name	ResourceReference: evacuation.1ST.dxf (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	-271.0
Width	544.0
y	-322.0
Height	642.0
Omniverse	
Usd binding type	None
Advanced	
Use antialiasing	false
Show name	false

Cad Layers:

Visible	Layer
true	Оси
true	Racking
true	Перекресток оборудование
true	Размеры по перег.
true	Стеллаж
true	рамка A2 к

Group: groupFloor2

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	

Name	Value
x	840.0
y	360.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

CAD Drawing: cad1

Name	Value
General	
Invert Default Colors	false
Fill color	white
File Name	ResourceReference: evacuation.2nd.dxf (Resolved: true)
Show at runtime	true
Lock	false
Public	true
Position and size	
x	-270.0
Width	523.0
y	-238.0
Height	461.0
Omniverse	
Usd binding type	None
Advanced	
Use antialiasing	false
Show name	false

Cad Layers:

Visible	Layer
true	0
true	FURN
true	Racking
true	Контуры
true	Перекресток оборудование
true	Размеры по перег.
true	Стеллаж
true	Стены АХП
true	конт пособн.
true	объекты
true	СТЕНЫ

Group: groupDinnerTable

Name	Value
------	-------

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Position and size	
x	-170.0
y	-90.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: groupSt1

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	480.0
y	150.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-55.0
dX	0.0
y	-30.0

Name	Value
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line1

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-50.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line2

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-45.0
dX	0.0
y	-30.0
dY	65.0

Name	Value
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line3

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-40.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line4

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-35.0
dX	0.0
y	-30.0
dY	65.0
z	0.0

Name	Value
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line5

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-30.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line6

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-25.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0

Name	Value
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line7

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-20.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line8

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-15.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0

Name	Value
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line9

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-10.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line10

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-5.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	

Name	Value
Usd binding type	None
Advanced	
Show name	false

Line: line11

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	0.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line12

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	5.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None

Name	Value
Advanced	
Show name	false

Line: line13

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	10.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line14

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	15.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	

Name	Value
Show name	false

Line: line15

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	20.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line16

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	25.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line17

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	30.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line18

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	30.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line19

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	35.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line31

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	40.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line32

Name	Value
General	

Name	Value
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	45.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line33

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	50.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line34

Name	Value
General	
Show at runtime	true

Name	Value
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	50.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line35

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	55.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line36

Name	Value
General	
Show at runtime	true
Lock	false

Name	Value
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	60.0
dX	0.0
y	-30.0
dY	65.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: groupSt2

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	1050.0
y	150.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line37

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid

Name	Value
Position and size	
x	-25.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line38

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-20.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line39

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	

Name	Value
x	-15.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line40

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-10.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line41

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-5.0

Name	Value
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line42

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	0.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line43

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	5.0
dX	0.0

Name	Value
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line44

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	10.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line45

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	15.0
dX	0.0
y	-28.0

Name	Value
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line46

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	20.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line47

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	25.0
dX	0.0
y	-28.0
dY	60.0

Name	Value
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line48

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	30.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line49

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	35.0
dX	0.0
y	-28.0
dY	60.0
z	0.0

Name	Value
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line50

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-30.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line51

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-35.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0

Name	Value
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line52

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	-40.0
dX	0.0
y	-28.0
dY	60.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: groupMainMenu

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	0.0
y	-20.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Rectangle: rectangle2

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Fill color	white
Line width	1.0
Line style	solid
Position and size	
x	0.0
Width	1100.0
y	20.0
Height	45.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Rectangle: rectangle9

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Fill color	new Color(230, 234, 239)
Line width	1.0
Line style	solid
Position and size	
x	0.0
Width	1100.0
y	60.0
Height	6.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line53

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	navy
Line width	1.0
Line style	solid
Position and size	
x	0.0
dX	1100.0
y	60.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: textHead

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Text	
Text	Hongik Evacuation model
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	12.0
y	25.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: group1

Name	Value
General	
Dynamic: Visible	!addExit
Show at runtime	true

Name	Value
Lock	false
Embedded icon	false
Public	true
Position and size	
x	30.0
y	160.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Polyline: polyline1

Name	Value
General	
Polyline closed	false
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	red
Line width	2.0
Line style	dashed
Position and size	
x	0.0
y	-20.0
z	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line88

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	red
Line width	2.0
Line style	solid
Position and size	

Name	Value
x	-7.0
dX	14.0
y	11.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line89

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Appearance	
Line color	red
Line width	2.0
Line style	solid
Position and size	
x	-7.0
dX	14.0
y	-20.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: group2

Name	Value
General	
Dynamic: Visible	alarm
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	495.0
y	735.0
z	0.0

Name	Value
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Polyline: polyline2

Name	Value
General	
Polyline closed	true
Show at runtime	true
Lock	false
Public	true
Appearance	
Fill color	new Color(27, 166, 7)
Line width	1.0
Line style	solid
Position and size	
x	-35.0
y	-15.0
z	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text10

Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Text	
Text	Exit
Appearance	
Color	white
Alignment	LEFT
Position and size	
x	-27.0
y	-14.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None

Name	Value
Advanced	
Show name	false

Group: group3

Name	Value
General	
Dynamic: Visible	alarm&&addExit
Show at runtime	true
Lock	false
Embedded icon	false
Public	true
Position and size	
x	57.0
y	155.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Polyline: polyline3

Name	Value
General	
Polyline closed	true
Show at runtime	true
Lock	false
Public	true
Appearance	
Fill color	new Color(27, 166, 7)
Line width	1.0
Line style	solid
Position and size	
x	50.0
y	-13.0
z	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text19

Name	Value
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Name	Value
General	
Show at runtime	true
Lock	false
Public	true
Text	
Text	Exit
Appearance	
Color	white
Alignment	LEFT
Position and size	
x	5.0
y	-13.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

View Area: viewArea

Name	Value
General	
Show name	false
Position and size	
x	0.0
Width	1114.0
y	0.0
Height	780.0

Check Box: checkbox

Name	Value
General	
Enabled	true
Link	addExit
Link	addExit
Link to	true
Lock	false
Public	true
Action	
Action	if (value) evacuation.exits.add(line30); else evacuation.exits.remove(line30);
Appearance	
Text color	black
Position and size	
x	200.0
Width	20.0
y	83.0

Name	Value
Height	17.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false
Embedded icon	false

Button: button

Name	Value
General	
Dynamic: Enable	!alarm
Enabled	true
Label text	Flood Alarm!
Lock	false
Public	true
Action	
Action	<pre> getEngine().setRealTimeMode(true); getEngine().setRealTimeScale(1); alarm = true; alarmBeginning = time(); checkbox.setEnabled(false); pedGoToCafeteria.cancelAll(); pedGoToWork.cancelAll(); pedGoToStairs.cancelAll(); pedGoToExit.cancelAll(); pedGoToTable.cancelAll(); pedWork.cancelAll(); pedWC.cancelAll(); pedEat.cancelAll(); pedSmoke.cancelAll(); visitorSource.set_limitArrivals(true); visitorSource.set_maxArrivals(visitorSource.countPeds()); workerSource.set_rate(0); evacuationStart.add(time(),0); evacuationStart.add(time(),getPeopleInsideCount()); </pre>
Appearance	
Text color	new Color(220, 0, 0)
Position and size	
x	720.0
Width	280.0
y	50.0
Height	60.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false
Embedded icon	false

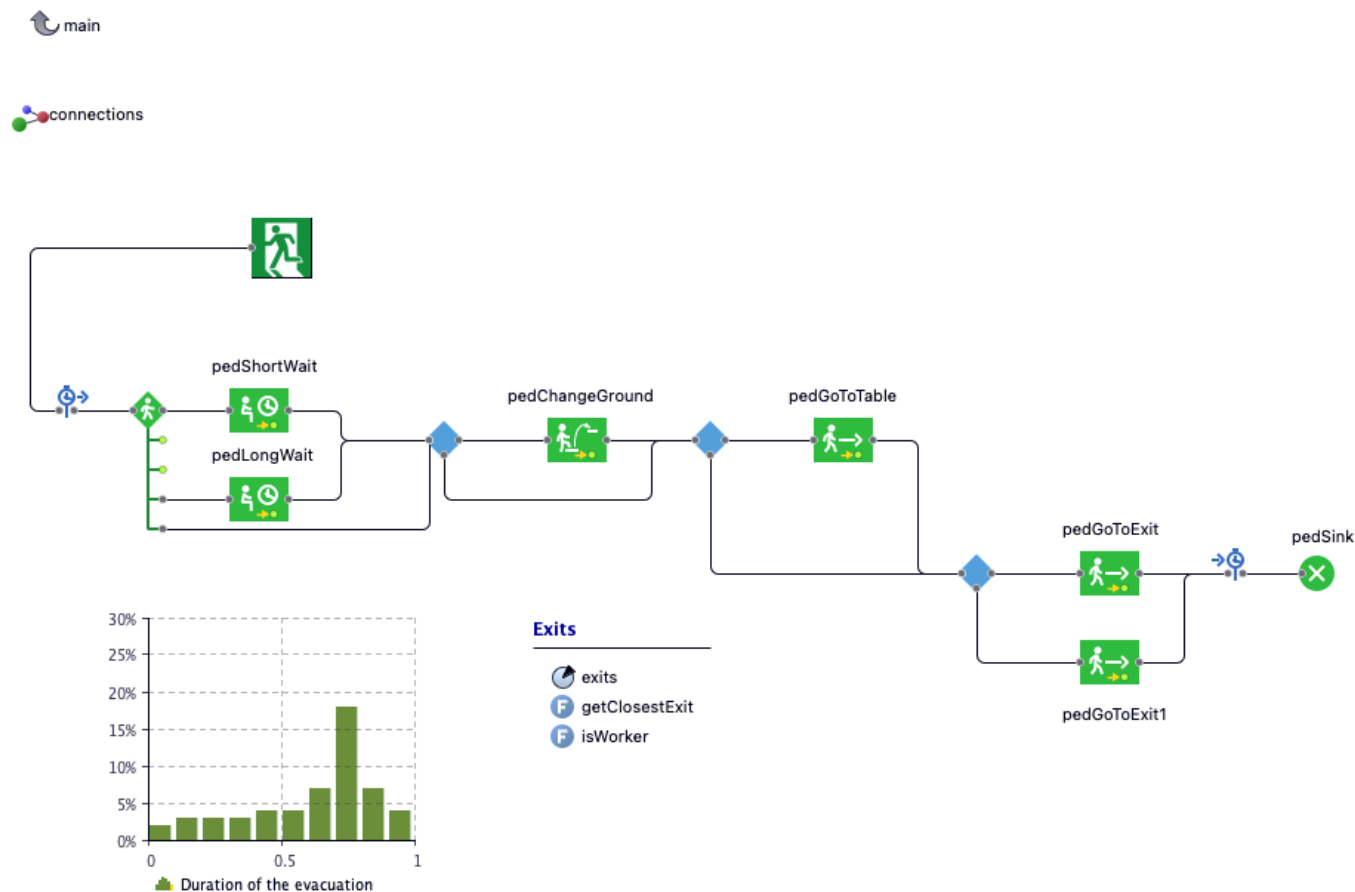
Link to agents: connections

Name	Value
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Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Evacuation

Name	Value
Agent in flowcharts	
Use in flowcharts as	Agent
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

Parameter: exits

Name	Value
General	
Array	false
Default value	new ArrayList<TargetLine>()
Type	ArrayList<TargetLine>
Show at runtime	true

Name	Value
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Function: getClosestExit

Name	Value
General	
Return type	TargetLine
Return type:	Returns value
Show at runtime	true
Show name	true
Function body	
Body	<pre>double dst = Double.POSITIVE_INFINITY; TargetLine result = null; for (TargetLine e : exits) if (getDistance(ped.getX(),ped.getY(), e.getX(), e.getY())< dst){ result = e; dst = getDistance(ped.getX(),ped.getY(), e.getX(), e.getY()); } return result;</pre>
Advanced	
Access type	default
System dynamics units	false

Arguments:

Name	Type
ped	Visitor

Function: isWorker

Name	Value
General	
Return type	boolean
Return type:	Returns value
Show at runtime	true
Show name	true
Function body	
Body	return ped instanceof OfficeWorker;
Advanced	
Access type	default
System dynamics units	false

Arguments:

Name	Type
ped	Agent

Port: port

Name	Value
General	
Show at runtime	true
Public	true
Show name	false

PedGoTo: pedGoToExit1

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_LINE
Target line	exits.get(1)
Reach tolerance	0.25

PedGoTo: pedGoToExit

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_LINE
Target line	exits.get(0)
Reach tolerance	0.25

SelectOutput: selectOutput2

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Select True output	false
Condition	isWorker(agent) && randomTrue(0.3)
Condition is stochastic or volatile	true

SelectOutput: selectOutput1

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Select True output	false
Condition	getClosestExit(agent).equals(exits.get(0))
Condition is stochastic or volatile	true

PedSink: pedSink

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true

Name	Value
Public	false
Logging	true

SelectOutput: selectOutput

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Select True output	false
Condition	agent.getLevel().equals(main.secondGround)
Condition is stochastic or volatile	true

TimeMeasureStart: timeMeasureStart

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

TimeMeasureEnd: timeMeasureEnd

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false

Name	Value
Logging	true

Agent Parameters:

Name	Value
TimeMeasureStart blocks	{timeMeasureStart}
Dataset capacity	100

PedGoTo: pedGoToTable

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Mode	self.MODE_REACH_TARGET
Target	self.LOCATION_POINT
X	((OfficeWorker)ped).table.getX()
Y	((OfficeWorker)ped).table.getY()
Reach tolerance	0.25

PedSelectOutput: pedSelectOutput1

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute: Visitor]
Show name	false
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Use:	self.TYPE_CONDITIONS
Condition 1	randomTrue(0.3)
Condition 2	false
Condition 3	false

Agent Parameters:

Name	Value
Condition 4	randomTrue(0.1)
Choice is stochastic or volatile	true
On enter	//ped.setDesiredSpeed(2*ped.getDesiredSpeed()); //ped.setSpeed(2*ped.getSpeed());

PedWait: pedShortWait

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Waiting location	self.LOCATION_POINT
X	ped.getX()
Y	ped.getY()
Delay ends	self.DELAY_TIMEOUT
Delay time	triangular(0,1,3)
Group members leave together	false
Delay starts when	self.DELAY_STARTS_SINCE_BEGIN_WAIT
Maximum capacity	true
Reach tolerance	0.25

PedWait: pedLongWait

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Waiting location	self.LOCATION_POINT

Agent Parameters:

Name	Value
X	ped.getX()
Y	ped.getY()
Delay ends	self.DELAY_TIMEOUT
Delay time	triangular(1,2,5)
Group members leave together	false
Delay starts when	self.DELAY_STARTS_SINCE_BEGIN_WAIT
Maximum capacity	true
Reach tolerance	0.25

PedChangeLevel: pedChangeGround

Name	Value
General	
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Population of agents	false
Generic Parameters Substitutes	[Generic Parameter Substitute:]
Show name	true
Advanced	
Show at runtime	true
Public	false
Logging	true

Agent Parameters:

Name	Value
Exit line on current level	main.line21
Entry line on new level	main.line20
Reach tolerance	0.25

Histogram: chart

Name	Value
General	
Show mean	false
Show CDF	false
Show PDF	true
Lock	false
Public	true
Data update	
Analysis auto update	false
Appearance	
Bars relative width	0.8
Labels vertical position	DEFAULT
Labels Text Color	darkGray
Chart Area Grid Color	darkGray
Position and size	
x	80.0

Name	Value
Width	240.0
y	280.0
Height	220.0
Legend	
Show legend	true
Legend size	30.0
Legend text color	black
Chart area	
Chart Area: X Offset	40.0
Chart Area: Width	180.0
Chart Area: Y Offset	20.0
Chart Area: Height	150.0
Chart Area: Background Color	white
Chart area border color	black
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Histogram Data Items:

Title	Dataset	PDF Color	CDF Color	Mean Color	Width	Low % Color	High % Color
Duration of the evacuation	timeMeasureEnd.distribution	oliveDrab	slateBlue	mediumOrchid	1	oliveDrab	gold

Rectangle: rectangle

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	true
Public	true
Appearance	
Fill color	white
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	190.0
Width	40.0
y	30.0
Height	40.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only

Name	Value
On click	return true;
Show name	false

Image: image

Name	Value
General	
Show at runtime	true
Lock	false
Embedded icon	true
Public	true
Position and size	
x	190.0
Width	40.0
y	30.0
Height	40.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Line: line87

Name	Value
General	
Show at runtime	true
Lock	false
Public	false
Appearance	
Line color	black
Line width	1.0
Line style	solid
Position and size	
x	380.0
dX	120.0
y	320.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Text: text17

Name	Value
General	
Show at runtime	true
Lock	false
Public	false
Text	
Text	Exits
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	380.0
y	300.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show in	2D only
Show name	false

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: OfficeWorker

Name	Value
Agent actions	
Startup code	person.setVisible(false);
Agent in flowcharts	
Use in flowcharts as	Pedestrian
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None

Name	Value
Advanced	
Extends other agent	ClassReference: evacuation.Visitor (Resolved: true)
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTIES	- Recurring Event Properties
Limit the number of data samples	false

 connections

 table

 isSmoker

 isHungry

 worktime

Scale: scale

Name	Value
General	
Inherited from parent	true
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0

Parameter: table

Name	Value
General	
Array	false
Default value	null
Type	Attractor
Show at runtime	true
Show name	true
Value editor	
Editor control	Text
Advanced	
System dynamics units	false
Save in snapshot	true

Variable: isSmoker

Name	Value
General	
Type	boolean
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: isHungry

Name	Value
General	
Initial value	true
Type	boolean
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: worktime

Name	Value
General	
Type	double
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

3D Object: officeWorker

Name	Value
General	
Scale	2.0
Auto scale	false
File Name	ResourceReference: evacuation.3d/officeworker.dae (Resolved: true)
Show at runtime	true

Name	Value
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Agent Type: Visitor

Name	Value
Agent in flowcharts	
Use in flowcharts as	Pedestrian
Dimensions and movement	
Speed	(10 : MPS)
Rotate animation towards movement	true
Rotate vertically (incline on ramps)	false
Space and network	
Space Type	Continuous
Layout Type	User-defined
Network type	User-defined
Enable steps	false
Advanced Java	
Generic	false
Omniverse	
Usd binding type	None
Advanced	
Flowchart block	false
Logging	true
Auto-create datasets	true
AOC_DATASETS_UPDATE_TIME_PROPERTY	- Recurring Event Properties

Name	Value
ES	
Limit the number of data samples	false



Scale: scale

Name	Value
General	
Unit	meters
Scale	10.0
Type	Defined graphically
Length, pixels	100.0
Show at runtime	false
Lock	false
Public	false
Position and size	
x	0.0
y	-150.0
Rotation	0.0

3D Object: person

Name	Value
General	
Scale	2.0
Auto scale	false
File Name	ResourceReference: evacuation.3d/person4.dae (Resolved: true)
Show at runtime	false
Lock	false
Public	true
Position and size	
x	0.0
y	0.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Usd ignore generated object	false
Advanced	
Show in	2D and 3D
Show name	false
Internal lights	Off

Link to agents: connections

Name	Value
General	
Show at runtime	true
Show name	true
Communication	
Message type	Object
Animation	
Draw line	false

Simulation Experiment: Simulation

Description: There is a two-storey building in this model. On the first floor, of this 24 person office building, the workers work and socialize. The restroom is also located on this floor.
Workers typical work day consists of eight hours.

On the second floor there is a cafeteria, where workers and outside customers have lunch.
(The number of outside customers usually depends what time of day it is - there will be more during lunch than in the morning for breakfast)

There is only one stairway between the two floors and one or two exits from the first floor.

When the fire alarm is activated everyone in the building will go to one of the exits.
Some may not react immediately and keep working while others may freeze or panic.
Some may also want to get back to their tables before exiting the building.

The alarm will start at the specified time or by pressing the "Fire alarm" button in the simulation window.

Name	Value
General	
Bypass Initial Simulation Screen	false
Maximum available memory	128
Agent type	Main
Model time	
Execution mode	Real time with scale
Real time scale	1.0
Stop option	Stop at specified date
Initial time	0.0
Initial date	Fri Mar 14 08:00:00 GMT 2025
Final date	Fri Mar 14 22:00:00 GMT 2025
Randomness	
Random Number Generation Type	Random seed (unique simulation runs)
Window	
Title	Evacuation : Simulation
Enable zoom and panning	false
Enable developer panel	true
Show developer panel on start	false
Advanced	
Load root from snapshot	false
Description	
Description	There is a two-storey building in this model. On the first floor, of this 24 person office building, the workers work and socialize. The restroom is also located on this floor.

Name	Value
	Workers typical work day consists of eight hours. On the second floor there is a cafeteria, where workers and outside customers have lunch. (The number of outside customers usually depends what time of day it is - there will be more during lunch than in the morning for breakfast) There is only one stairway between the two floors and one or two exits from the first floor. When the fire alarm is activated everyone in the building will go to one of the exits. Some may not react immediately and keep working while others may freeze or panic. Some may also want to get back to their tables before exiting the building. The alarm will start at the specified time or by pressing the "Fire alarm" button in the simulation window.

Hongik Evacuation model



서초구 2층 업무용 건물 홍수 발생 시나리오

📍 건물 개요

1층: 사무실(약 24명 근무), 화장실 위치 | 2층: 구내식당 (직원 및 외부 손님 이용)

출입구: 1층 외부 출구 2개 | 이동 경로: 1층-2층 연결 계단 1개

△□ 화재 발생 상황

갑작스러운 폭우로 건물 주변 침수 심화 → 홍수 경보 시스템 작동

홍수 경보 울림 → 건물 내 인원 대피 시작

♂ 대피 반응 (개인별 행동 차이)

✓ 즉각 대피: 신속하게 출구로 이동

✓ 인지 지연: 상황 파악 못하고 업무 지속

✓ 혼란 발생: 공황 상태로 움직이지 못함

✓ 물건 회수 시도: 자리로 복귀해 소지품 챙김

□□ 구내식당(2층) 대피 상황

직원 및 외부 손님 계단 이용 증가 → 병목 현상 발생

일부는 공황 상태로 혼란 가중

■ 출구 대피 및 관리 대응

2개 출구로 대피 진행 → 혼잡 증가 | 안전 관리자 도착 → 질서 유지 시도

침수 심화 → 불안감 증가 → 대피 속도 지연

🔍 시뮬레이션 적용

'홍수 경보' 버튼 클릭 또는 특정 시간 경과 후 자동 발생

시뮬레이션 분석 가능 항목:

건물 내 인원 행동 패턴 | 대피 속도 및 동선 평가

대피 경로 효율성 | 공황 및 혼란 영향 분석

이를 통해 재난 대응 방안 최적화 및 효과적인 대피 전략 수립 가능



Alarm time (minutes):

Variable: alarmTime

Name	Value
General	
Initial value	540
Type	double
Show at runtime	true
Show name	true
Advanced	

Name	Value
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Variable: m

Name	Value
General	
Type	MonteCarlo
Show at runtime	true
Show name	true
Advanced	
Access type	public
Constant	false
Save in snapshot	true
System dynamics units	false

Text: text1

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Alarm time (minutes):
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	720.0
y	770.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Image: image2

Name	Value
General	
Show at runtime	true
Lock	false
Position and size	
x	720.0
Width	490.0

Name	Value
y	260.0
Height	500.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text4

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	2 1: (24), 2: () : 1 2 : 1-2 1 → → () : : : : (2) → 2 → → → → '' :
Appearance	

Name	Value
Color	black
Alignment	LEFT
Position and size	
x	10.0
y	60.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Group: groupTitleBar

Name	Value
General	
Show at runtime	true
Lock	false
Position and size	
x	0.0
y	-20.0
Dynamic: y	-20
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Rectangle: rectangle2

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Fill color	white
Line width	1.0
Line style	solid
Position and size	
x	0.0
Width	1100.0
y	20.0
Height	45.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	

Name	Value
Usd binding type	None
Advanced	
Show name	false

Rectangle: rectangle9

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Fill color	new Color(230, 234, 239)
Line width	1.0
Line style	solid
Position and size	
x	0.0
Width	1100.0
y	60.0
Height	6.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line53

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Line color	navy
Line width	1.0
Line style	solid
Position and size	
x	0.0
dX	1100.0
y	60.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: textHead

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Hongik Evacuation model
Appearance	
Color	navy
Alignment	LEFT
Position and size	
x	12.0
y	25.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Edit Box: editbox

Name	Value
General	
Dynamic: Enable	true
Enabled	true
Maximum value	720
Minimum value	0
Link	alarmTime
Link	alarmTime
Link to	true
Lock	false
Appearance	
Fill color	white
Text color	black
Position and size	
x	950.0
Width	100.0
y	770.0
Height	21.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Parameter Variation Experiment: MonteCarlo

Name	Value
------	-------

Name	Value
General	
Maximum available memory	512
Agent type	Main
Model time	
Stop option	Stop at specified date
Initial time	0.0
Initial date	Mon Jul 23 08:00:00 GMT 2012
Final date	Mon Jul 23 22:00:00 GMT 2012
Randomness	
Random Number Generation Type	Random seed (unique simulation runs)
Replications	
Use replications	false
Window	
Title	Evacuation : MonteCarlo
Enable zoom and panning	false
Enable developer panel	true
Show developer panel on start	false
Advanced	
Allow parallel evaluations	true
Load root from snapshot	false

Parameter Variation Experiment Parameters:

Parameter	Value
alarmTime	uniform(0,8*hour())
addExit	randomTrue(0.5)
visitorArrivalMultiplier	
workerArrivalRate	

Evacuation : MonteCarlo

Iterations completed: ?

Parameters

Robustness ?

Rapidity ?

Redundancy ?

Resourcefulness ?

Text: text

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Evacuation : MonteCarlo

Name	Value
Appearance	
Color	royalBlue
Alignment	LEFT
Position and size	
x	40.0
y	30.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text1

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Iterations completed:
Appearance	
Color	new Color(70, 72, 74)
Alignment	LEFT
Position and size	
x	50.0
y	80.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text2

Name	Value
General	
Dynamic: Visible	getCurrentIteration() > 0
Show at runtime	true
Lock	false
Text	
Text	?
Dynamic: Text	format(getNumberOfCompletedIterations())
Appearance	
Color	darkSlateBlue
Alignment	RIGHT
Position and size	
x	240.0

Name	Value
y	80.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text3

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Parameters
Appearance	
Color	darkGray
Alignment	LEFT
Position and size	
x	50.0
y	110.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Line: line

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Line color	darkGray
Line width	2.0
Line style	solid
Position and size	
x	40.0
dX	210.0
y	130.0
dY	0.0
z	0.0
dZ	0.0
Z-Height	10.0
Omniverse	
Usd binding type	None

Name	Value
Advanced	
Show name	false

Text: text4

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Robustness
Appearance	
Color	new Color(70, 72, 74)
Alignment	LEFT
Position and size	
x	50.0
y	150.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text5

Name	Value
General	
Dynamic: Visible	getCurrentIteration() > 0
Show at runtime	true
Lock	false
Text	
Text	?
Dynamic: Text	format(alarmTime)
Appearance	
Color	darkSlateBlue
Alignment	RIGHT
Position and size	
x	240.0
y	150.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Rectangle: rectangle

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Fill color	whiteSmoke
Line width	1.0
Line style	solid
Position and size	
x	40.0
Width	210.0
y	170.0
Height	30.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text6

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Rapidity
Appearance	
Color	new Color(70, 72, 74)
Alignment	LEFT
Position and size	
x	50.0
y	180.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text7

Name	Value
General	
Dynamic: Visible	getCurrentIteration() > 0
Show at runtime	true
Lock	false
Text	

Name	Value
Text	?
Dynamic: Text	format(addExit)
Appearance	
Color	darkSlateBlue
Alignment	RIGHT
Position and size	
x	240.0
y	180.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text8

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Redundancy
Appearance	
Color	new Color(70, 72, 74)
Alignment	LEFT
Position and size	
x	50.0
y	210.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text9

Name	Value
General	
Dynamic: Visible	getCurrentIteration() > 0
Show at runtime	true
Lock	false
Text	
Text	?
Dynamic: Text	format(visitorArrivalMultiplier)
Appearance	
Color	darkSlateBlue
Alignment	RIGHT

Name	Value
Position and size	
x	240.0
y	210.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Rectangle: rectangle1

Name	Value
General	
Show at runtime	true
Lock	false
Appearance	
Fill color	whiteSmoke
Line width	1.0
Line style	solid
Position and size	
x	40.0
Width	210.0
y	230.0
Height	30.0
z	0.0
Z-Height	10.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text10

Name	Value
General	
Show at runtime	true
Lock	false
Text	
Text	Resourcefulness
Appearance	
Color	new Color(70, 72, 74)
Alignment	LEFT
Position and size	
x	50.0
y	240.0
z	0.0
Rotation	0.0

Name	Value
Omniverse	
Usd binding type	None
Advanced	
Show name	false

Text: text11

Name	Value
General	
Dynamic: Visible	getCurrentIteration() > 0
Show at runtime	true
Lock	false
Text	
Text	?
Dynamic: Text	format(workerArrivalRate)
Appearance	
Color	darkSlateBlue
Alignment	RIGHT
Position and size	
x	240.0
y	240.0
z	0.0
Rotation	0.0
Omniverse	
Usd binding type	None
Advanced	
Show name	false